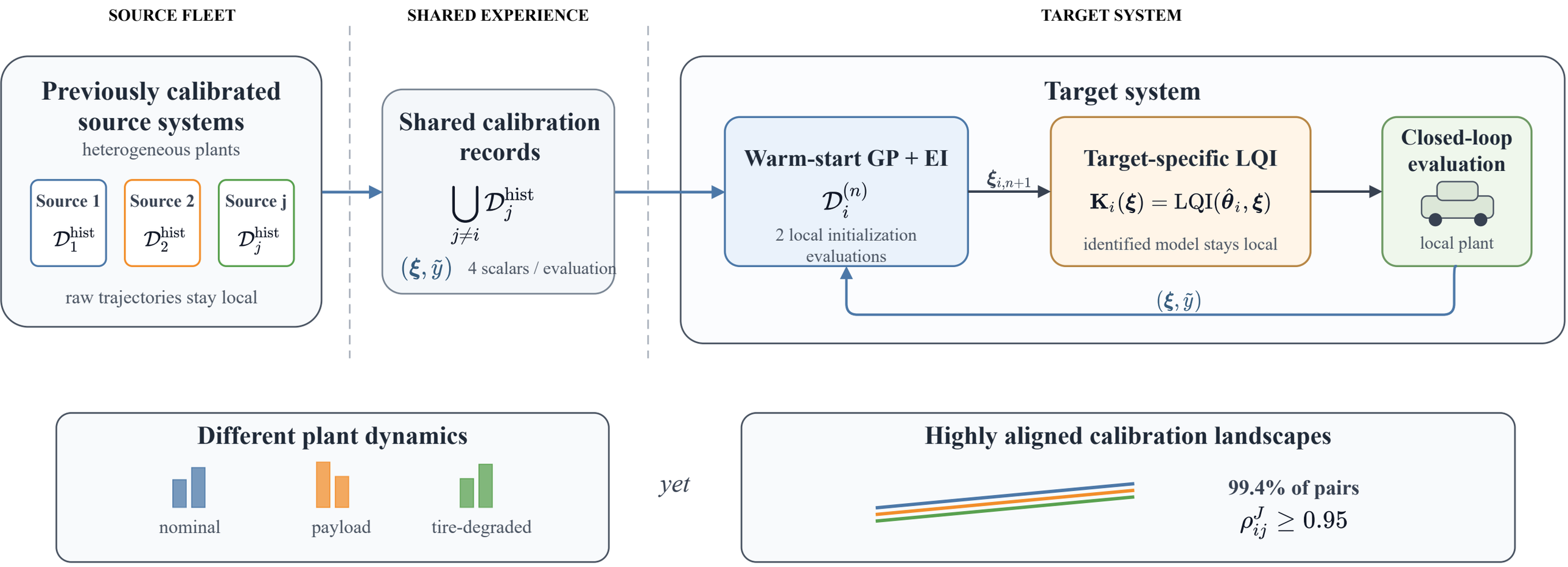

SOURCE FLEET
SHARED EXPERIENCE
TARGET SYSTEM
Previously calibrated source systems
heterogeneous plants
Source 1
$\mathcal{D}_1^{\text{hist}}$
Source 2
$\mathcal{D}_2^{\text{hist}}$
Source j
$\mathcal{D}_j^{\text{hist}}$
raw trajectories stay local
Shared calibration records
$\bigcup_{j \neq i} \mathcal{D}_j^{\text{hist}}$
$(\boldsymbol{\xi}, \tilde{y})$ 4 scalars / evaluation
Target system
Warm-start GP + EI
$\mathcal{D}_i^{(n)}$
2 local initialization evaluations
$\boldsymbol{\xi}_{i,n+1}$
Target-specific LQI
$\mathbf{K}_i(\boldsymbol{\xi}) = \text{LQI}(\hat{\boldsymbol{\theta}}_i, \boldsymbol{\xi})$
identified model stays local
Closed-loop evaluation
local plant
$(\boldsymbol{\xi}, \tilde{y})$
Different plant dynamics
nominal
payload
tire-degraded
yet
Highly aligned calibration landscapes
99.4% of pairs
$\rho_{ij}^J \geq 0.95$